Q1

Consider a uniaxial single lamina of HS-carbon/epoxy, with a 60 vol.% fibre content.
Assuming that the Young's modulus of HS carbon and epoxy are respectively 450 GPa and 3.5 GPa:

Calculate the Young's Modulus of the composite along the fibre axis (E_{C1}).

$$E_{C1} = E_f V_f + E_m (1 - V_f) = 271.4 \text{ GPa}$$

Q2

Consider a uniaxial single lamina of HS-carbon/epoxy, with a 60 vol.% fibre content.
Assuming that the Young's modulus of HS carbon and epoxy are respectively 450 GPa and 3.5 GPa:

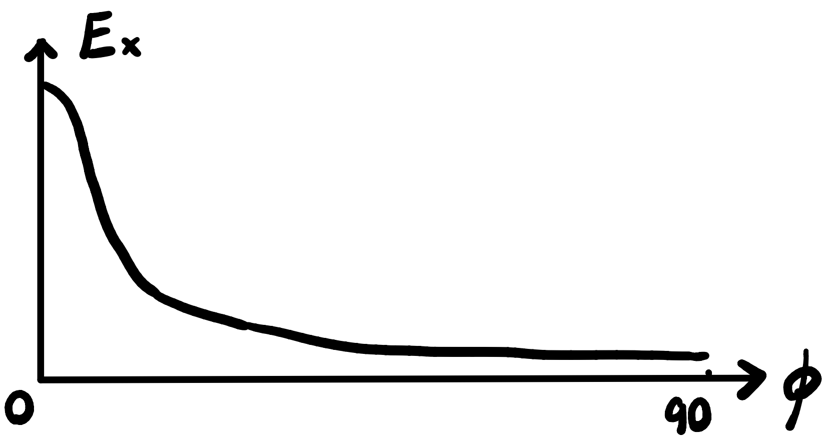
Calculate the Young's Modulus of the composite transverse to the fibre axis (E_{C2}).

$$\frac{1}{E_{c2}} = \frac{V_f}{E_f} + \frac{1-V_f}{E_m}$$

$$\therefore E_{c2} = 8.65 \text{ GPa}$$

Q3

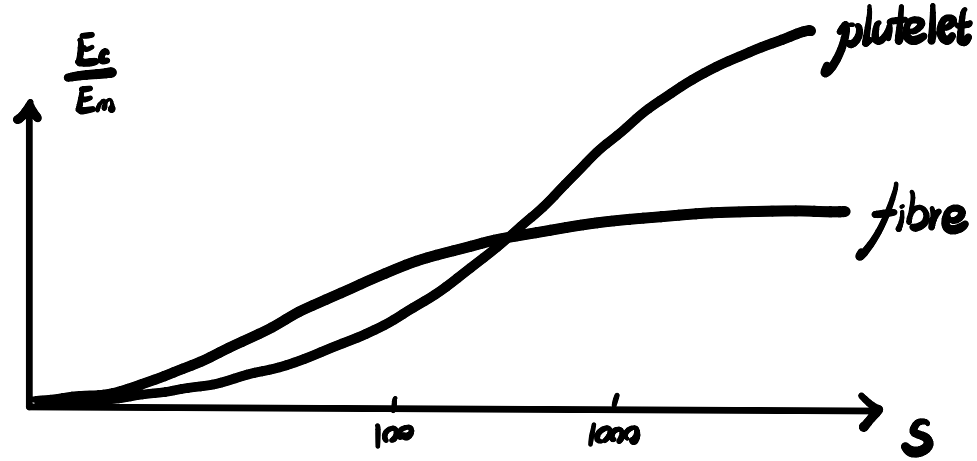
Draw an approximate plot of the composite modulus (Ex), as a function of the tilt angle ϕ, between ϕ = 0° (x-axis along the fibre axis), ϕ = 90° (x-axis transverse to the fibre axis).



Q4

Cite a micromechanical model that describes the variation of the Young's modulus of a composite as a function of the aspect ratio of the reinforcing fibre, and write the corresponding equation.

Halpin - Tsai model



where $E = \dfrac{E_m(1+\xi\eta V_f)}{1-\eta V_f}$ where $\eta = \dfrac{\frac{E_f}{E_m}-1}{\frac{E_f}{E_m}+\xi}$

Q5

Select the best composite material (fibre and matrix) and manufacturing method, for the following applications, briefly justifying your answer:

Decking for a large cruise ship

glass + epoxy

non-crimp fabrics + RIFT

Tennis racket

carbon + epoxy

bradied

Skin for a cargo container on civil aircraft

aramid + epoxy

2D woven

Q6

Select the best composite material (fibre and matrix) and manufacturing method, for the following applications, briefly justifying your answer:

Wind turbine blade

glass + epoxy (with wood / carbon / ... inside)

vacuum infusion

Hull of a yacht

carbon + epoxy

autoclave

Toilet Cover

no composite

Lower drag brace for the landing gear of a fighter aircraft

carbon + epoxy

uniaxial prepeg tapes